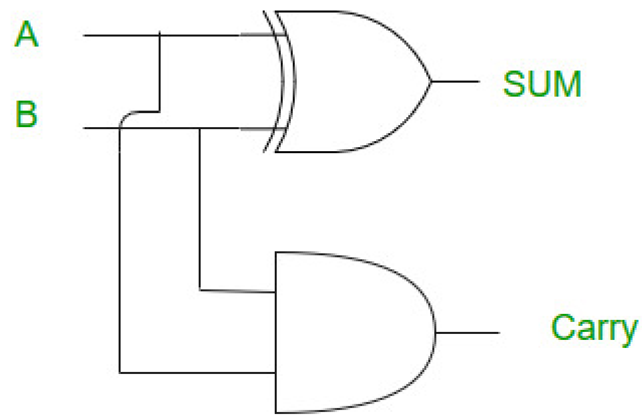
Half Adder

A half adder is a basic combinational circuit that adds two single-bit binary inputs (A and B) to produce a SUM using an XOR gate and a CARRY using an AND gate, without considering any carry-in from a previous stage.

- Performs binary addition of two single-bit inputs, generating a SUM ($A \oplus B$) and CARRY ($A \cdot B$).
- Cannot handle carry-in from a previous stage, making it suitable only for the first stage of multi-bit addition.



Truth Table of Half Adder

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Logical Expression of Half Adder

The Half Adder performs two operations, Sum and Carry. Therefore, two K-maps are used to derive their Boolean expressions.

Sum :

$$\text{Sum} = \text{A XOR B}$$

Carry :

$$\text{Carry} = \text{A AND B}$$

Advantages

- **Simple Design:** A half adder has a simple circuit design that uses only two logic gates (XOR and AND), making it easy to implement and understand.
- **Fast Operation:** Since it contains only a few logic gates, it produces the Sum and Carry outputs with very little delay.
- **Low Hardware Requirement:** It requires fewer hardware components, reducing circuit complexity and implementation cost.

Disadvantages

- **No Carry Input:** A half adder cannot accept a carry input from a previous stage, limiting its use in larger arithmetic operations.
- **Limited Functionality:** It can add only two single-bit binary numbers and cannot perform multi-bit addition by itself.
- **Requires Full Adders:** For multi-bit binary addition, half adders must be combined with full adders to handle carry propagation.

Applications

- **Binary Addition:** It is used to add two single-bit binary numbers and generate the corresponding Sum and Carry outputs.
- **Building Block for Full Adders:** Half adders are commonly used as the basic building blocks in the design of full adders.
- **Digital Arithmetic Circuits:** They are used in arithmetic units, calculators, processors, and other digital systems where simple binary addition is required.

Full Adder

Full Adder is a combinational circuit that adds three inputs and produces two outputs. The first two inputs are A and B and the third input is an input carry as C-IN. The output carry is designated as C-OUT and the normal output is designated as S which is SUM.

- The C-OUT is also known as the majority 1's detector, whose output goes high when more than one input is high.
- A full adder logic is designed in such a manner that can take eight inputs together to create a byte-wide adder and cascade the carry bit from one adder to another.
- We use a full adder because when a carry-in bit is available, another 1-bit adder must be used since a 1-bit half-adder does not take a carry-in bit.
- A 1-bit full adder adds three operands and generates 2-bit results.

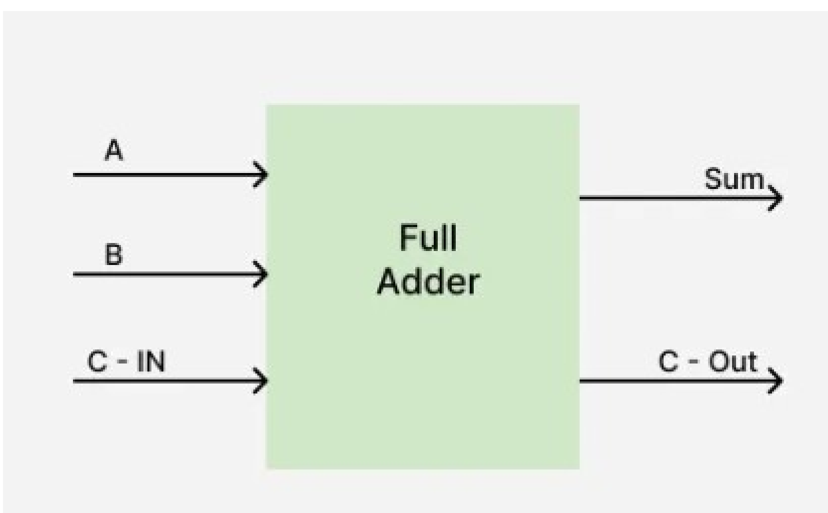
Full Adder Truth Table

A Full Adder takes three binary inputs:

- A (first bit)
- B (second bit)
- C-IN (carry input)

And it produces two outputs:

- Sum (S)
- Carry Out (C-OUT)



Here's the truth table for the full adder:

INPUT			OUTPUT	
A	**B**	**C-IN**	**Sum**	**C-OUT**
0	**0**	**0**	**0**	**0**
0	**0**	**1**	**1**	**0**
0	**1**	**0**	**1**	**0**
0	**1**	**1**	**0**	**1**
1	**0**	**0**	**1**	**0**
1	**0**	**1**	**0**	**1**
1	**1**	**0**	**0**	**1**
1	**1**	**1**	**1**	**1**

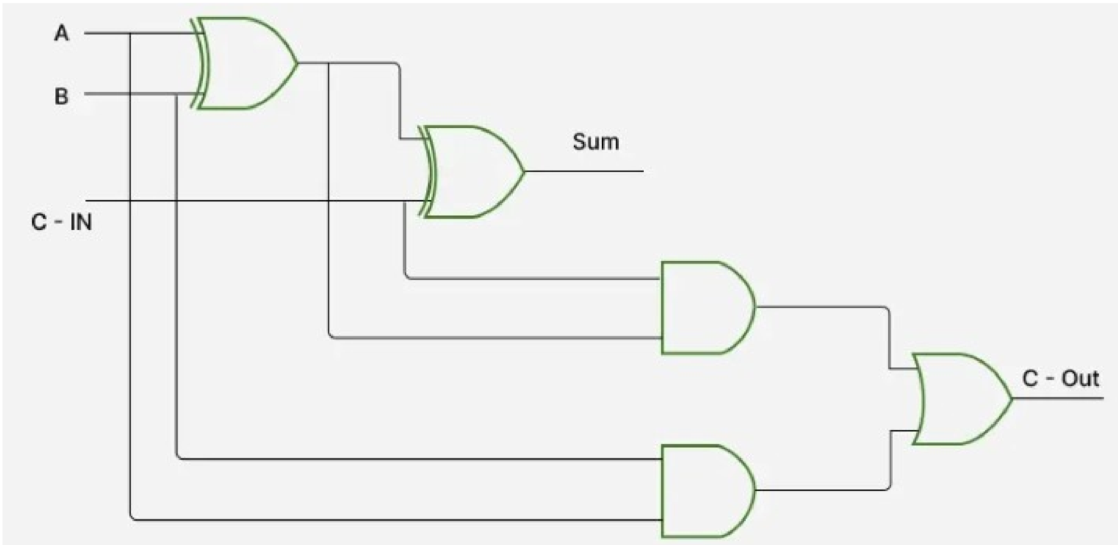
Logic Circuit of Full Adder

To implement a Full Adder using basic logic gates:

Sum (S) is implemented using XOR gates:

Use two XOR gates:

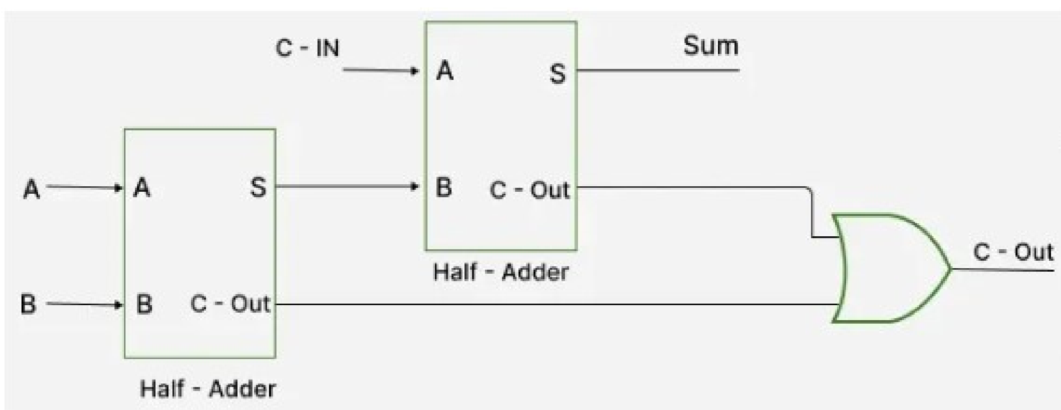
- First XOR gate: $A \oplus B$
- Second XOR gate: $(A \oplus B) \oplus$ C-IN to get the final sum S.



Sum: $S = A \oplus B \oplus C\text{-}IN$

$C\text{-}OUT = A\,B + C\text{-}IN\,(A \oplus B)$

Implementation of Full Adder using Half Adders



Ring Counter

A ring counter is a typical application of the Shift register. The ring counter is almost the same as the shift counter. The only change is that the output of the last flip-flop is connected to the input of the first flip-flop in the case of the ring counter but in the case of the shift register it is taken as output. Except for this, all the other things are the same.

Ring Counter

It is a type of digital counter used in circuits, typically built with flip-flops. It works by shifting a 1 through a series of flip-flops, one at a time, in a loop. It's called a ring counter because the 1 loops back to the start once it reaches the end, forming a cycle.

No. of states in Ring counter = No. of flip-flop used

Key Characteristics of a Ring Counter

- **Single bit '1'**: Only one bit in the counter is set to 1 at any given time.
- **Cyclic Behavior**: The pattern of the bits follows a cyclic behavior and repeats after a fixed number of steps.
- **No Reset**: The ring counter doesn't automatically reset to zero. Instead, it starts at a defined state and then continues cycling.

Working Of Ring Counter

- A Ring Counter is typically built using flip-flops (D, T, or JK flip-flops).
- It consists of n flip-flops, where n is the number of states in the counter. The number of flip-flops determines the number of unique states the counter will cycle through.
- One bit in the counter is set to 1, and the rest of the bits are set to 0. In each clock cycle, the 1 bit shifts through the flip-flops, and the pattern repeats.
- The output of the last flip-flop is fed back into the first flip-flop, forming a loop, hence the name Ring Counter.

Example of a 4-bit Ring Counter

0000

JK →

Let's consider a 4-bit Ring Counter:

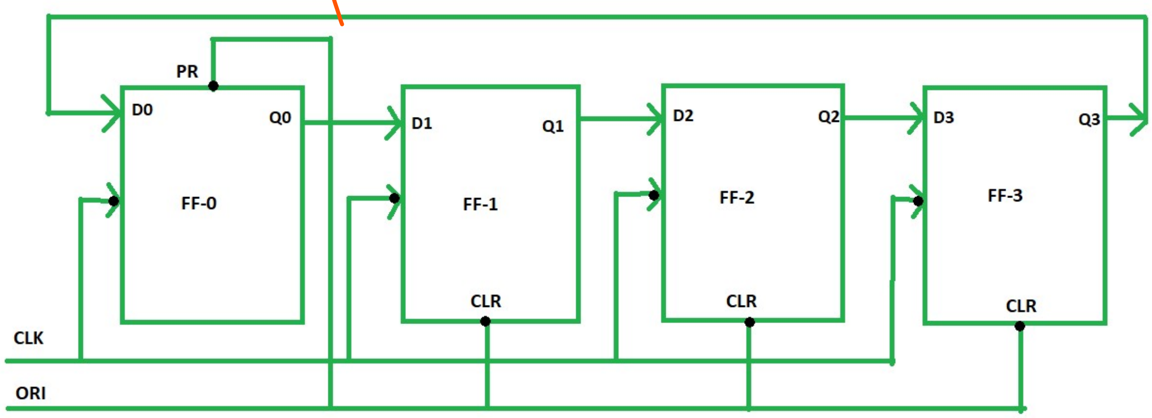
1. **Initial state**: 1000 (the first flip-flop is set to 1, and others are set to 0).
2. **Clock cycle 1**: The '1' bit shifts to the next flip-flop, resulting in the state 0100.
3. **Clock cycle 2**: The '1' bit shifts to the next flip-flop, resulting in the state 0010.

4. **Clock cycle 3**: The '1' bit shifts to the next flip-flop, resulting in the state 0001.

5. **Clock cycle 4**: The '1' bit shifts back to the first flip-flop, and the cycle repeats, resulting in the state 1000.

State Sequence:

1000 → 0100 → 0010 → 0001 → 1000



Ring Counter

Advantages of Ring Counter

- **Simplicity**: It's easy to design, especially for small applications.
- **Low Resource Use**: You don't need a lot of components compared to other counters.
- **Predictable Output**: It will always go through the same set of states in the same order.
- **Easy to Decode:** Each state is distinct and easy to decode because it uses one-hot encoding only one output is high at a time.
- **Glitch-Free Output:** Since only one flip-flop is active at a time, there's less chance of output glitches, making it more reliable in certain applications.

4

Disadvantages of Ring Counter

- **Limited States**: You can only cycle through as many states as the number of flip-flops in the counter. So, for a 4-bit counter, you can only have four unique states.
- **Not for Complex Patterns**: It's not great if you need a counter that counts in random or non-binary patterns since it's limited to the cyclic pattern of shifting one 1 through the flip-flops.

- **Needs Initialization:** A ring counter must be properly initialized (usually with only one flip-flop set to 1) before it starts working. If not, it can get stuck in an invalid state.
- **Hardware Cost Increases:** As you add more flip-flops to increase the number of states, the hardware needed also increases, which can make the design bulky for large state requirements.

Applications of Ring Counter

Ring counters are used when you need a system that goes through a specific, repeating set of states. For example:

- **Sequence Generation**: It's useful when you want a predictable sequence, like in simple control systems or timing circuits.
- **Control Systems**: Ring counters are great for things like multiplexers or encoder circuits, where the order of operations needs to follow a certain cycle.
- **Simple Design:** Ring counters are easy to design and understand because each state has only one flip-flop set to 1, making debugging and analysis straightforward.
- **Finite State Machines (FSM):** Used in systems that move through a known set of states in a loop, such as vending machines or elevator controllers.
- **Shift Register Based Applications:** Ring counters are a special type of shift register, so they are used where shift registers are needed for data movement.

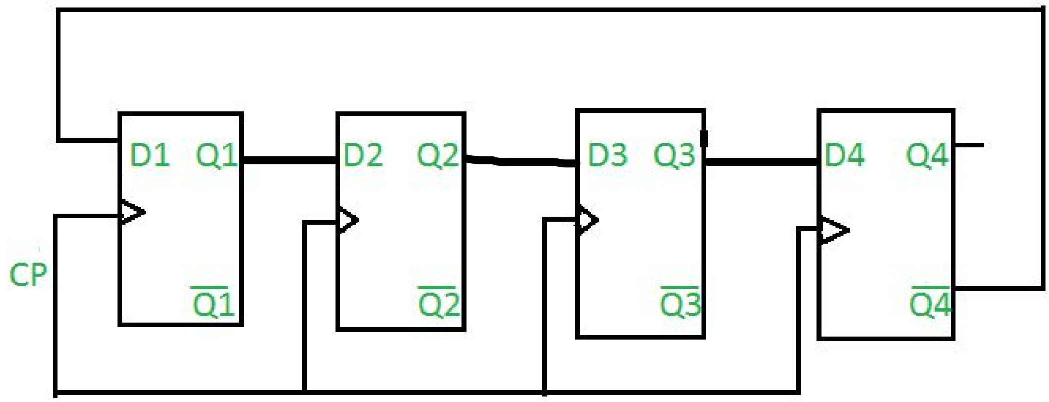
n-bit Johnson Counter

JA Johnson counter, also known as a twisted ring or creeping counter, is a synchronous shift register counter where the complemented output of the last flip-flop is fed into the input of the first. It uses *n* flip-flops to generate *2n* unique states, offering better state efficiency than regular ring counters.

- Requires *n* flip-flops to generate *2n* distinct states, making it more efficient than a ring counter.
- Commonly used for applications needing self-decodable outputs and efficient state utilization.

Total number of used and unused states in n-bit Johnson counter:
number of used states=2n
number of unused states=2^n *- 2*n*

Example:
If n=4
4-bit Johnson counter:



Truth Table

CP	Q1	Q2	Q3	Q4
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

Advantages

- The Johnson counter has same number of flip flop but it can count twice the number of states the ring counter can count.
- It can be implemented using D and JK flip flop.
- Johnson ring counter is used to count the data in a continuous loop.
- Johnson counter is a self-decoding circuit.

Disadvantages

- Johnson counter doesn't count in a binary sequence.
- In Johnson counter more number of states remain unutilized than the number of states being utilized.

- The number of flip flops needed is one half the number of timing signals.
- It can be constructed for any number of timing sequence.

Applications

- Johnson counter is used as a synchronous decade counter or divider circuit.
- It is used in hardware logic design to create complicated Finite states machine. ex: ASIC and FPGA design.
- The 3 stage Johnson counter is used as a 3 phase square wave generator which produces 1200 phase shift.
- It is used to divide the frequency of the clock signal by varying their feedback.

Ring Counter vs Johnson Counter

Parameters	Ring Counter	Johnson Counter
Configuration	A ring counter employs the carry-in of the last flip-flop into the input of the first flip-flop without any manipulation.	In Johnson counter, the complement of output of the last flip-flop is applied to the input of the first flip-flop.
Number of Flip-Flops	'n' flip-flops are required to count 'n' states.	'n' flip-flops are required to count '2n' states.
Counting Sequence	It counts in a simple binary sequence often having one '1' and the rest '0's in each state.	It counts in a twisted sequence, where the output is a mixture of binary 1s and 0s.
Number of States	It can Generate 'n' unique states	It can Generate '2n' unique states
Unused states	None, because all the states are utilized	'2n-2n' states are unused
Self-Decoding Capability	Its not self-decoding since additional circuitry is needed	Its self-decoding makes it simpler for certain applications

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Circuit Complexity	Since it does not require inversion feedback, thus the circuit is simple	Due to inversion, the circuit is slightly more complex

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